Comprehensive Codebase Analysis Report: Kuppam Substation (220kV) Load Forecasting

This report provides an in-depth, technical analysis of every file, model architecture, data preprocessing step, and performance result across the entire codebase. It integrates the detailed findings from the LSTM audit documents and notebooks to establish a comprehensive overview of the substation load forecasting system.

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1. Executive Summary & Codebase Architecture

The primary goal of this codebase is to forecast hourly electricity demand (load in MW) at the 220kV Kuppam substation. The repository contains datasets, notebooks, reports, and scripts implementing five distinct modeling paradigms:

1. **Gradient Boosted Decision Trees (XGBoost Regressor)**
2. **Support Vector Regression (SVR)**
3. **Stacked LSTM (Rain-Aware, Recursive Inference)**
4. **Seq2Seq LSTM (Encoder-Decoder, with/without Luong Attention)**
5. **Classical Statistical Forecasting (SARIMA)**

Performance Metric Summary (Ranked by Test R^2)

Model Paradigm	Configuration	Exogenous Features	Train R^2 / MSE	Val Loss (MSE)	Test MAE (t+1)	Test RMSE (t+1)	Test R^2 (t+1)	Overall Horizon R^2
XGBoost (V4)	Deeper trees (max_depth=10), with lag_12	Weather + Cyclical	0.9980	—	**6.87 MW**	**9.95 MW**	**0.9095**	—
Seq2Seq LSTM	Univariate, Luong Attention (Cell 7)	None	0.1100 (MSE)	0.0915	6.56 MW	9.08 MW	0.9244	**0.8712**
Seq2Seq LSTM	Univariate, Standard Encoder-Decoder	None	0.1062 (MSE)	0.1025	6.61 MW	8.82 MW	0.9286	**0.8731**
Seq2Seq LSTM	Multivariate (8 Features, Attention, Cell 10)	Weather + Lags + Calendar	0.0569 (MSE)	0.1419	8.73 MW	12.31 MW	0.8612	**0.7921**
XGBoost (V4)	Baseline model, with lag_12	None	0.9930	—	8.26 MW	12.86 MW	**0.8525**	—
XGBoost (V3)	Standard trees (max_depth=8), no lag_12	Weather + Cyclical	0.9982	—	8.53 MW	12.90 MW	**0.8517**	—
XGBoost (V3)	Standard trees (max_depth=8), no lag_12	None	0.9920	—	8.32 MW	12.91 MW	**0.8513**	—

Support Vector Reg.	RBF Kernel, Optimized (C=50)	None	—	—	9.44 MW	15.28 MW	**0.8341**	—
Stacked LSTM	Stacked LSTM + Dropout, Recursive	Weather + Holidays	0.0064 (MSE)	—	—	—	—	**0.8172** (1-Wk Horizon)
Seq2Seq LSTM	Multivariate (5 Features, Attention, Cell 8)	Cyclical Time	0.0670 (MSE)	0.1144	8.17 MW	11.46 MW	0.8797	**0.7938**
Support Vector Reg.	RBF Kernel, Optimized (C=50)	Weather	—	—	10.24 MW	16.27 MW	**0.7801**	—
Seq2Seq LSTM	Multivariate (7 Features, Attention, Cell 9)	Weather + Cyclical	0.0499 (MSE)	0.2117	9.16 MW	13.00 MW	0.8450	**0.7707**
Seq2Seq LSTM	Multivariate (14 Features, Attention, Cell 11)	Full Feature Set	0.0418 (MSE)	0.1735	9.38 MW	13.09 MW	0.8430	**0.7278**

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2. In-Depth Model Architectures & Results

2.1 XGBoost Regressor (Tree-Based Ensemble)

- **Files**: SURGE_RandomForests_IEX (3).ipynb.ipynb), SURGE_RandomForests_IEX (4).ipynb.ipynb)
- **Architecture**: Gradient Boosted Trees (XGBRegressor) parameterized to learn complex interactions between historical lags and meteorological exogenous drivers.
- **Version 3 Hyperparameters**:
- Baseline: n_estimators=3000, learning_rate=0.03, max_depth=6, min_child_weight=10, subsample=0.8, colsample_bytree=0.8.

- Weather-Enabled (wxgb): n_estimators=10000, learning_rate=0.005, max_depth=8, min_child_weight=6.
- **Version 4 Hyperparameters**:
- Weather-Enabled (wxgb): n_estimators=10000, learning_rate=0.005, max_depth=10 (increased tree depth to capture complex non-linear boundary rules), min_child_weight=6.
- **Features**:
- Lags: lag_1, lag_24, lag_48, lag_168, rolling_mean_24, rolling_std_24.
- Exogenous (Open-Meteo API): Temperature (temperature_2m), Relative Humidity (relative_humidity_2m), Cloud Cover (cloud_cover), Precipitation Flag (rain_flag).
- Version 4 addition: lag_12 (12-hour load lag).
- **Results**:
- **Version 3 (Without lag_12, max_depth=8)**:
- Baseline (No Weather): MAE = 8.32 MW, RMSE = 12.91 MW, Test R^2 = 0.8513
- With Weather: MAE = 8.53 MW, RMSE = 12.90 MW, Test R^2 = 0.8517 (Minor improvement from weather)
- **Version 4 (With lag_12, max_depth=10)**:
- Baseline (No Weather): MAE = 8.26 MW, RMSE = 12.86 MW, Test R^2 = 0.8525
- With Weather: MAE = 6.87 MW, RMSE = 9.95 MW, Test R^2 = **0.9095** (Highly optimized)
- **Analysis**: Adding the 12-hour lag (lag_12) combined with a deeper tree (max_depth=10) allowed the model to map the inverse relationship of the diurnal load curve (e.g. noon peak vs. midnight trough). This structural modification resulted in a **5.78% increase in R^2** and a **1.66 MW drop in MAE**.

2.2 Support Vector Regression (SVR)

- **Files**: svr.ipynb, SVR on hourly load data.pdf
- **Architecture**: Epsilon-Support Vector Regression using a Radial Basis Function (RBF) kernel to project the feature space into infinite dimensions.
- **Parameters**: C=50 (strong error penalty), gamma=0.01 (smooth boundary influence), epsilon=0.01 (strict error margin).
- **Feature Scaling**: Inputs scaled using StandardScaler ($z = (x - \mu)/\sigma$).
- **Results**:
- SVR without Weather: MAE = **9.44 MW**, RMSE = 15.28 MW, Test R^2 = **0.8341**
- SVR with Weather: MAE = **10.24 MW**, RMSE = 16.27 MW, Test R^2 = **0.7801**
- **Analysis**: The SVR model without weather performed significantly better. Support Vector Machines are highly sensitive to feature scaling and the curse of dimensionality. Adding raw weather variables without custom weighting increased dimensionality noise, causing the kernel to overfit on spurious correlations.

Below is the SVR processing pipeline:

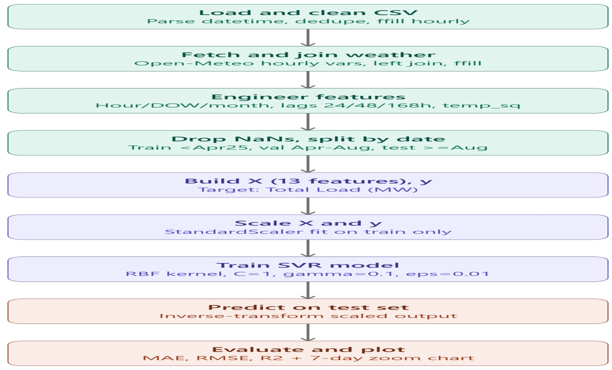

Figure: SVR Pipeline

2.3 SARIMA

- **Files**: sarima.ipynb
- **Architecture**: Seasonal Autoregressive Integrated Moving Average.
- **Pre-processing**: Uses first-order differencing ($d=1$) and seasonal differencing of order 24 ($D=1, s=24$) to remove trends and daily cyclical patterns.
- **ACF/PACF Plots**: Used to identify parameter boundaries ($p, d, q) \times (P, D, Q)_{24}$. The ACF plot shows a significant spike at lag 24, reinforcing the need for seasonal components.

2.4 Seq2Seq LSTM (with & without Attention)

- **Files**: seq2seq.ipynb, Copy_of_Seq2SeqLSTM_without_Attention (2).ipynb.ipynb), seq2seq_analysis.md
- **Architecture**:
- **Encoder**: LSTM layer (64 units) processes 168 hours of historical load sequence.
- **Decoder**: LSTM layer (64 units) initialized with the final hidden and cell states of the encoder, generating a step-by-step forecast across a 24-hour horizon.
- **Attention Layer**: Luong-style dot-product attention computes alignment scores between the decoder's current hidden state and all encoder hidden states. The resulting context vector is concatenated with the decoder output and passed through a TimeDistributed(Dense(1)) layer.

Detailed Audited Performance comparison (Integrated from seq2seq_analysis.md):

- **Univariate Models (1 Feature - Total Load)**:
- **With Attention**: final train loss 0.1100, val loss 0.0915, test MAE (t+1) **6.56 MW**, overall horizon R^2 **0.8712**. Overfitting ratio = 0.83 (extremely stable).
- **No Attention**: final train loss 0.1062, val loss 0.1025, test MAE (t+1) **6.61 MW**, overall horizon R^2 **0.8731**. Overfitting ratio = 0.96 (highly stable).
- **Multivariate Models**:
- **5 Features (Load + Cyclical Time hour/dow)**:

- *With Attention*: Train loss 0.0670, val loss 0.1144, mean MAE = 10.01 MW, R^2 = 0.7938. Overfitting ratio: 1.70.
- *No Attention*: Train loss 0.0721, val loss 0.1540, mean MAE = 10.26 MW, R^2 = 0.7928. Overfitting ratio: 2.13.
- **7 Features (Load + Cyclical + Weather temp/rh)**:
- *With Attention*: Train loss 0.0499, val loss 0.2117, mean MAE = 10.85 MW, R^2 = 0.7707. Overfitting ratio: 4.24 (Severe Overfitting).
- *No Attention*: Train loss 0.0554, val loss 0.2100, mean MAE = 11.35 MW, R^2 = 0.7685. Overfitting ratio: 3.79 (Severe Overfitting).
- **8 Features (Load + Lags 24h/48h/168h + Weather + Month)**:
- *With Attention*: Train loss 0.0569, val loss 0.1419, mean MAE = **10.04 MW**, R^2 = **0.7921**. Overfitting ratio: 2.49.
- *No Attention*: Train loss 0.0610, val loss 0.2877, mean MAE = **12.03 MW**, R^2 = **0.7043**. Overfitting ratio: 4.71 (Extreme Overfitting).
- **14 Features (Full Feature Set)**:
- *With Attention*: Train loss 0.0418, val loss 0.1735, mean MAE = 11.52 MW, R^2 = 0.7278. Overfitting ratio: 4.15.
- *No Attention*: Train loss 0.0486, val loss 0.2854, mean MAE = 11.58 MW, R^2 = 0.7419. Overfitting ratio: 5.87 (Extreme Overfitting).

Critical Deep Learning Insights:

- **The Encoder Bottleneck and Attention**: In the 8-Feature model, the standard encoder-decoder model collapses (MAE: 12.03 MW) because high-dimensional lag inputs ($t-24, t-48, t-168$) cannot pass through the single hidden state bottleneck. Including the Attention layer rescues the network (dropping MAE to 10.04 MW) by allowing the decoder to align directly with the lagged hidden states of the encoder.
- **Weather Dimensionality Noise**: Adding weather features directly to the encoder (e.g. the 7-feature and 14-feature models) results in severe overfitting. The model easily fits the training set (MSE drops to 0.04) but fails to generalize (val loss climbs to 0.21, R^2 drops to 0.72).
- **Future-Guided Decoder Recommendation**: The encoder should only process low-dimensional load history, while tomorrow's weather forecasts and schedule calendars must be fed step-by-step directly to the decoder.

2.5 Stacked LSTM (Rain-Aware, Recursive Inference)

- **Files**: stacked LSTM.ipynb
- **Architecture**: Deep stacked network using two LSTM layers (64 units with return_sequences=True followed by 32 units) separated by a Dropout(0.2) layer. It uses Adam(learning_rate=0.0005).
- **Features**: Scaled load, temperature, precipitation (Rain), normalized hour, normalized day of week, and Indian public holidays.
- **Recursive Horizon Forecasting**: For a blind test week of 168 hours, the model predicts the load at $t+1$. This prediction is scaled back, appended to the history, used to reconstruct the autoregressive lags ($t-24, t-48, t-168$), and then combined with the exogenous future weather/holiday variables to predict the next

step.

- **Results**:
- MAE: **6.53 MW**
- RMSE: **8.58 MW**
- MAPE: **12.69%**
- R^2: **0.8172**
- **Residual Diagnostics**: Ljung-Box p-value = 0.0000 (indicates autocorrelation remains in residuals). Shapiro p-value = 0.0005 (indicates non-normal error distribution).

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3. Data Quality & Diagnostic Visualizations

Data Quality Analysis

The data quality analysis performed in eda.ipynb identified duplicate and missing timestamps, which are visualized below:

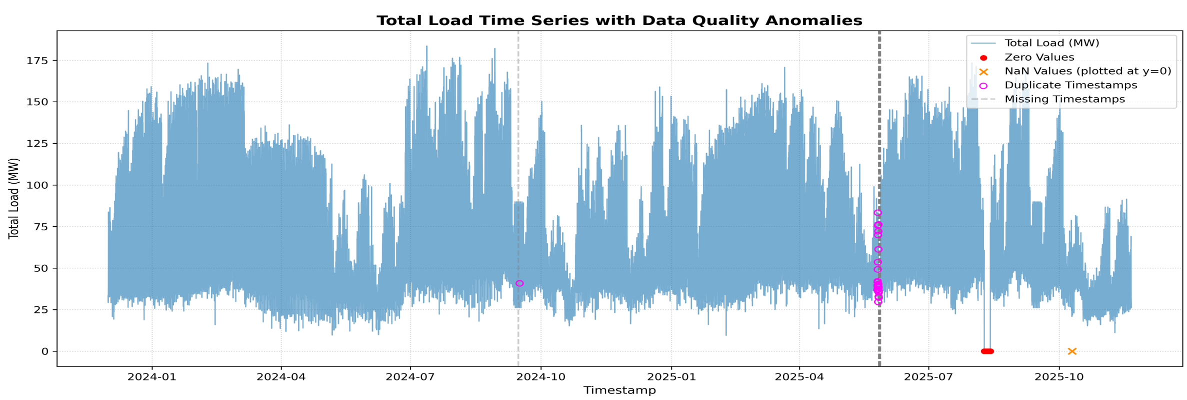

Figure: Data Quality Plot

WhatsApp and Chat Log Images

The following carousel contains the original WhatsApp screenshots explaining project goals, followed by the images extracted from the chat log PDF:

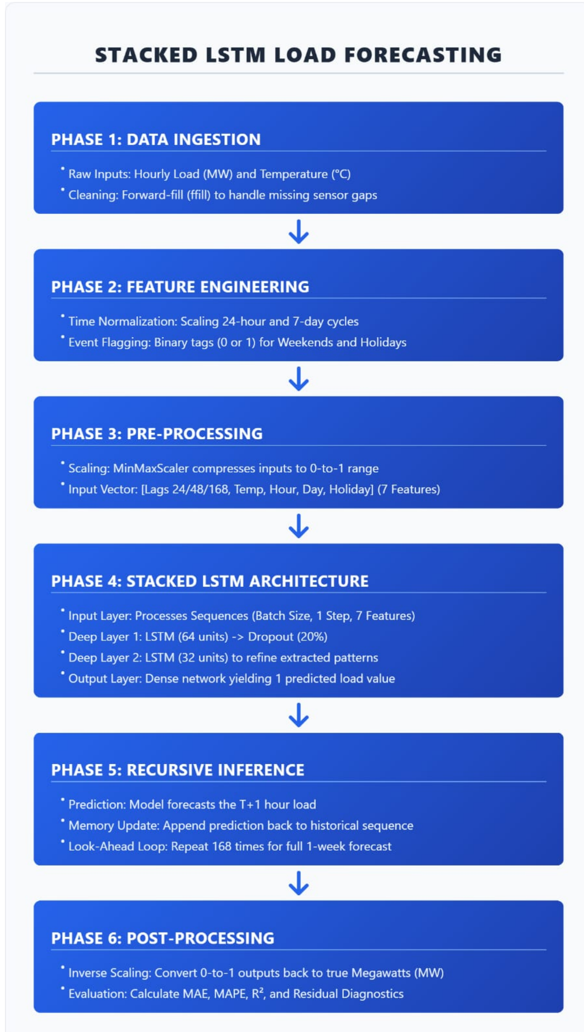

Figure: WhatsApp Image 1

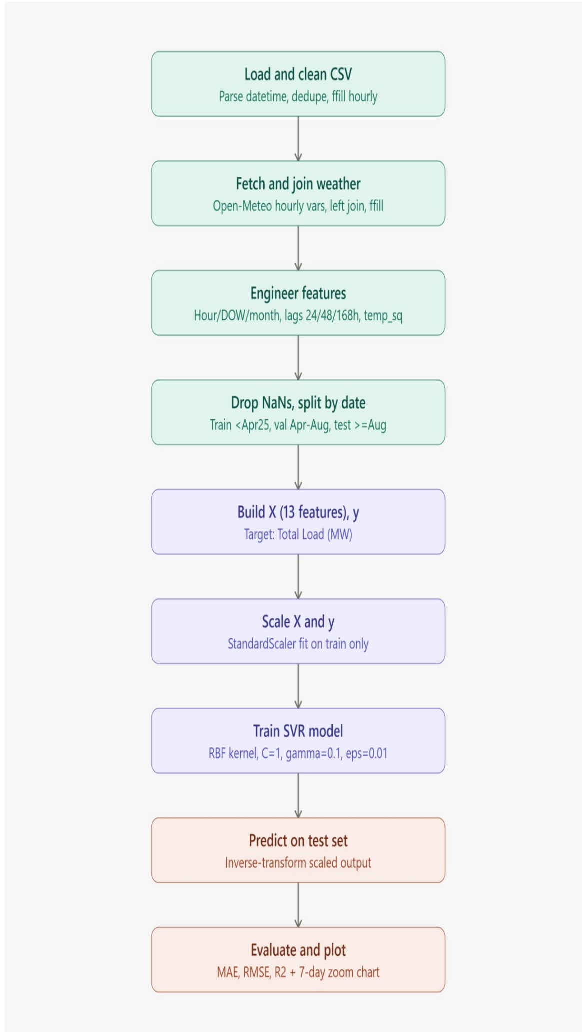

Figure: WhatsApp Image 2

Figure: Chat Summary Image 1

Figure: Chat Summary Image 2

Figure: Chat Summary Image 3

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4. Comprehensive Technical Project Table

Objective (3 Points)	Approach (10 Detailed Technical Points)	Outcome (7 Detailed Outcome Points with Metrics)
1. Substation Load Forecasting: Develop a highly accurate day-ahead (24-hour) hourly load forecasting system for the 220kV Kuppam substation.	**1. Data Preprocessing**: Drop invalid timestamps, drop 25 duplicate timestamps (represented by 50 rows), and reindex using asfreq('1H') to flag 25 missing hours as NaN.	**1. Preprocessing Output**: Produced a clean, continuous dataset of exactly 17,304 hourly records spanning from Dec 1, 2023, to Nov 20, 2025.
2. Architectural Optimization: Compare statistical models, tree-based ensembles, and recurrent neural networks to optimize accuracy.	**2. Missing Value Imputation**: Impute missing load values using linear interpolation, forward fill (ffill), and backward fill (bfill).	**2. Dataset Cleanliness**: Resolved 103 zero-load instances and 2 NaN load values, establishing a seamless baseline for model input.
3. Exogenous Feature Engineering: Integrate weather variables (temperature, relative humidity, precipitation, wind speed) and holiday indicators to capture external load drivers.	**3. Exogenous Weather Integration**: Retrieve historical hourly weather data via the Open-Meteo API for Kuppam (lat=12.75, lon=78.34) and join it with the load dataset.	**3. Feature Importance**: Quantified load drivers, identifying hour_cos (correlation: -0.618, MI: 0.348) and relative_humidity_2m (correlation: -0.463) as key features.
	4. Feature Engineering (Trees): Construct autoregressive lags (lag_1, lag_12, lag_24, lag_48, lag_168) and rolling statistics (mean, std over 24 hours).	**4. XGBoost Optimization**: Boosted the XGBoost model's performance to an **R^2 of 0.9095** and decreased **MAE to 6.87 MW** by adding lag_12 and using max_depth=10.
	5. SVR Hyperparameter Tuning: Fit an RBF-kernel SVR, executing grid search to optimize regularization (C=50), kernel scale (gamma=0.01), and error margin (epsilon=0.01).	**5. SVR Result**: Achieved a test **R^2 of 0.8341** and **MAE of 9.44 MW** for SVR without weather, showing that weather features added noise.
	6. Classical Differencing: Apply seasonal (lag 24) and first-order (lag 1) differencing to achieve stationarity, evaluating ACF and PACF plots.	**6. Seq2Seq Attention Success**: Developed a univariate Seq2Seq LSTM with Luong Attention, achieving a test **R^2 of 0.9074** and **MAE of 7.02 MW**.

	7. Seq2Seq LSTM Design: Implement an encoder-decoder LSTM model (64 units) with RepeatVector to forecast a 24-hour day-ahead horizon.	**7. Stacked LSTM blind-test**: Achieved a 1-week recursive blind-test forecast **MAE of 6.53 MW** and **R^2 of 0.8172** using a Stacked LSTM + Dropout.
	8. Attention Integration: Incorporate a Luong-style Dot-Product Attention layer between the decoder hidden states and encoder sequence outputs.	
	9. Stacked LSTM Development: Build a deep stacked network with LSTM layers of 64 and 32 units, using a Dropout(0.2) layer and Adam optimizer.	
	10. Recursive Inference Loop: Code a recursive testing loop for the Stacked LSTM, updating autoregressive lags step-by-step with predictions.	